

Savitribai Phule Pune University
T.Y.B.Sc. (Cyber and Digital Science)
CDS-308-MJP Practical based on CDS-307-MJ Web and Network
Security Fundamentals
Time : 3 Hours **Max.Marks:35**

Q.1) Configure a small network with two or more systems and assign appropriate IP addresses.

Verify the configuration. [15M]

Q.2) Use the ping command to test connectivity between two systems and explain the results. [15M]

Q.3) Viva [5M]

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Q.1) Use traceroute/tracert to identify the path followed by packets to a destination. [15M]

Q.2) Configure a basic LAN and troubleshoot a connectivity problem caused by an incorrect IP address or subnet mask. [15M]

Q.3) Viva [5M]

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Q.1) Perform a basic Nmap scan on an authorized system and identify the active hosts. [15M]

Q.2) Scan an authorized host to identify open TCP ports and record the results. [15M]

Q.3) Viva [5M]

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- Q.1) Identify the services running on open ports using appropriate Nmap options. [15M]
- Q.2) Compare the results of a basic Nmap scan and a service/version detection scan. [15M]
- Q.3) Viva [5M]

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Q.1) Analyze an Nmap scan result and prepare a table containing port number, protocol, state, and detected service. [15M]

Q.2) Capture network traffic using Wireshark and identify different protocols present in the capture. [15M]

Q.3) Viva [5M]

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Q.1) Apply a Wireshark filter to display only ICMP packets and analyze a ping communication. [15M]

Q.2) Capture DNS traffic and identify the source, destination, query, and response information. [15M]

Q.3) Viva [5M]

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Q.1) Analyze an HTTP packet capture and identify important fields such as source IP, destination IP, method, and response status. [15M]

Q.2) Examine a packet capture and identify traffic that appears unusual or suspicious. Give reasons for your observation. [15M]

Q.3) Viva [5M]

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- Q.1) Configure Burp Suite and a browser in an authorized lab environment to intercept HTTP requests. [15M]
- Q.2) Capture a web application's GET request using Burp Suite and identify its important components. [15M]
- Q.3) Viva [5M]

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- Q.1) Capture a POST request and identify the parameters and headers transmitted to the server. [15M]
- Q.2) Use Nikto on an authorized test web application and identify the security-related findings reported by the tool. [15M]
- Q.3) Viva [5M]

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- Q.1) Analyze an intercepted web request and identify possible input-validation or security weaknesses without exploiting the system. [15M]
- Q.2) Configure basic firewall rules to allow required network traffic and block unnecessary traffic. [15M]
- Q.3) Viva [5M]

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Q.1) Create a password policy requiring minimum password length and complexity, and verify that it is applied. [15M]

Q.2) Configure a system to restrict unnecessary network services and verify the changes. [15M]

Q.3) Viva [5M]

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Q.1) Perform a basic security audit of a system and document at least five security configuration improvements. [15M]

Q.2) Perform a vulnerability scan on an authorized lab system and classify the identified findings according to severity. [15M]

Q.3) Viva [5M]

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Q.1) Create a Report on vulnerability-scan findings and recommend suitable remediation steps for each finding. [15M]

Q.2) Prepare a security assessment report containing the target, tools used, methodology, findings, evidence, risk level, and recommended solutions. [15M]

Q.3) Viva [5M]

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Q.1) Configure two systems in a small LAN, verify their IP configuration, and demonstrate successful connectivity. [15M]

Q.2) Use ping and traceroute/tracert to verify connectivity and identify the route to a destination. [15M]

Q.3) Viva [5M]

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Q.1) Perform an Nmap scan on an authorized lab host and record all discovered open ports. [15M]

Q.2) Identify the services associated with the discovered ports and analyze the security significance of the results. [15M]

Q.3) Viva [5M]

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- Q.1) Capture traffic generated by a ping operation in Wireshark and identify the ICMP request and reply packets. [15M]
- Q.2) Capture DNS traffic and identify the DNS query and corresponding response. [15M]
- Q.3) Viva [5M]

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Q.1) Use Wireshark to analyze a sample network capture and identify at least three protocols. [15M]

Q.2) Identify any unusual or suspicious communication in the capture and provide evidence from the packet details. [15M]

Q.3) Viva [5M]

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Q.1) Configure Burp Suite to intercept traffic from an authorized web application and capture an HTTP request. [15M]

Q.2) Modify and resend a safe test request in the authorized lab and identify the request headers and parameters. [15M]

Q.3) Viva [5M]

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Q.1) Run Nikto against an authorized test web server and record the important findings. [15M]

Q.2) Classify the findings as configuration issues or potential vulnerabilities and suggest suitable remediation . [15M]

Q.3) Viva [5M]

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Q.1) Configure a basic host firewall to block an unnecessary port while allowing required traffic. [15M]

Q.2) Verify the firewall rule by testing the permitted and blocked connectivity. [15M]

Q.3) Viva [5M]

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- Q.1) Configure a password policy with suitable length and complexity requirements. [15M]
- Q.2) Create a report to test the password policy using sample passwords and identify which passwords are accepted and which are rejected [15M]
- Q.3) Viva [5M]

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Q.1) Perform a basic vulnerability scan on an authorized lab system. [15M]

Q.2) Prepare a table of identified vulnerabilities with severity, affected service, and recommended remediation. [15M]

Q.3) Viva [5M]

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Q.1) Prepare a complete security assessment report based on the vulnerability-scan results. [15M]

Q.2) Include target details, tools, methodology, findings, evidence, risk level, and remediation recommendations in the report. [15M]

Q.3) Viva [5M]

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Q.1) Set up a small authorized lab network and document its topology, IP addresses, and connectivity test results. [15M]

Q.2) Perform an Nmap scan of the lab network and summarize the active hosts, open ports, and detected services. [15M]

Q.3) Viva [5M]

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Q.1) Capture and analyze network traffic from the lab using Wireshark and identify normal and suspicious traffic patterns. [15M]

Q.2) Perform a basic authorized web-security assessment using Burp Suite or Nikto and document the observations. [15M]

Q.3) Viva [5M]